Malabsorption & Diseases of Small Bowel 2

Intended learning outcomes:

- By the end of this lecture the student will be able to:

1. Understand the common diseases of small bowel .
2. Know how to manage these diseases .

Small intestinal bacterial overgrowth (SIBO)(Blind loop syndrome)

Introduction

The normal duodenum and jejunum contain fewer than 104/mL organisms and these are usually derived from saliva. The count of coliform organisms never exceeds 103/mL.

In bacterial overgrowth, there may be 108 1010/mL organisms, most of which are normally found only in the colon.

Pathophysiology and Causes

The most important causes are:

1.Loss of gastric acidity

2.Impaired intestinal motility

3.Structural abnormalities that allow colonic bacteria to gain access to the small intestine.

Bacterial overgrowth can occur in patients with small bowel diverticuli or following surgery that creates ‘blind loops’ of bowel, e.g. Billroth II or Roux-en-Y reconstructions.

- **Diabetic autonomic neuropathy** reduces small bowel motility and affects enterocyte secretion.

- **In systemic sclerosis**, bacterial overgrowth arises because the circular and longitudinal layers of the intestinal muscle are fibrosed and motility is abnormal.

- **<u>In idiopathic hypogammaglobulinaemia</u>**, bacterial overgrowth occurs because the IgA and IgM levels in serum and jejunal secretions are reduced.

22.51 Causes of small bowel bacterial overgrowth

Mechanism	Examples
Hypo- or achlorhydria	Pernicious anaemia Partial gastrectomy Long-term PPI therapy
Impaired intestinal motility	Scleroderma Diabetic autonomic neuropathy Chronic intestinal pseudo-obstruction
Structural abnormalities	Gastric surgery (blind loop after Billroth II operation) Jejunal diverticulosis Enterocolic fistulae* Extensive small bowel resection Strictures*
Impaired immune function	Hypogammaglobulinaemia

*Most commonly caused by Crohn's disease.

Clinical Features

1. Patients commonly present with bloating, abdominal pain, distension and diarrhoea.
2. Nutritional deficiencies can occur in severe cases, e.g. vitamin B12 deficiency can arise as the bacteria utilise vitamin B12.
3. Other features of malabsorption.
4. Symptoms can commonly overlap with other diagnoses such as IBS and functional dyspepsia.
5. Symptoms of the underlying cause may be evident.

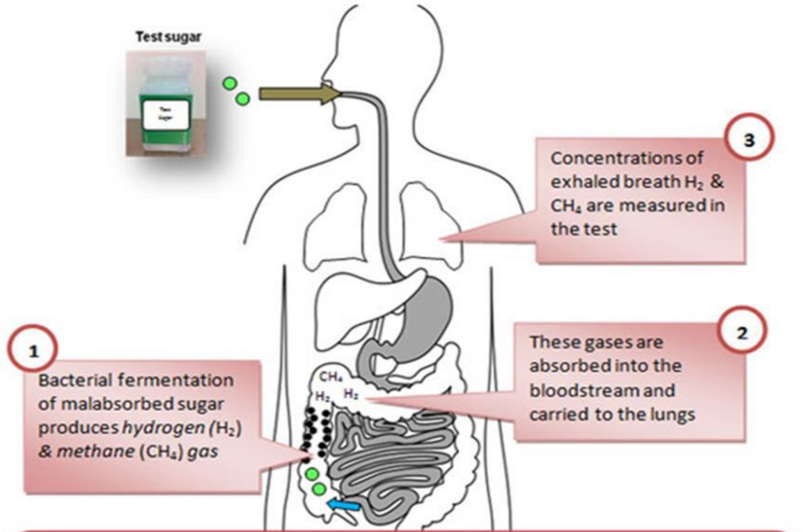
Investigations

-Culture of small bowel aspirate, obtained at endoscopy, is the **<u>gold standard</u>** for diagnosis , BUT?!

-Bacterial overgrowth can also be diagnosed non-invasively using hydrogen breath tests, although they lack sensitivity.

-Serum levels of vitamin B12 and folate.

Hydrogen breath test : measurement of breath samples for hydrogen after oral ingestion of 75 g glucose or 10 g lactulose. If bacteria are present within the small bowel, they rapidly metabolise the glucose/lactulose, causing an early rise in exhaled hydrogen.



Concentrations of breath hydrogen and methane are used to indicate if the test sugar is malabsorbed or if proximal bacterial overgrowth is present.

Management

1.Treatment of the underlying cause.

2.**Broad-spectrum antibiotic** for 2 weeks is the first line treatment. (e.g. Rifaximin , ciprofloxacin, metronidazole)

*Some pts. do not respond to Abi , that’s why they require up to 4 weeks of treatment and, in a few, continuous rotating courses of antibiotics are necessary.

3.Intramuscular **vitamin B12 supplementation** may be needed in chronic cases.

4.Anti diarrhoeal drugs .

Whipple Disease

Rare condition caused by infection with the Gram positive bacillus Tropheryma whipplei, which becomes resident within macrophages in the bowel mucosa.

Characterised by infiltration of small intestinal mucosa by ‘foamy’ macrophages, which stain positive with periodic acid–Schiff (PAS) reagent.

The disease is a multisystem one and almost any organ can be affected, sometimes long before gastrointestinal involvement becomes apparent.

23.45 Clinical features of Whipple’s disease

Gastrointestinal (> 70%)	
• Diarrhoea (75%)	• Protein-losing enteropathy
• Steatorrhoea	• Ascites
• Weight loss (90%)	• Hepatosplenomegaly (< 5%)
Musculoskeletal (65%)	
• Seronegative large joint arthropathy	• Sacroiliitis
Cardiac (10%)	
• Pericarditis	• Endocarditis
• Myocarditis	• Coronary arteritis
Neurological (10%–40%)	
• Apathy	• Myoclonus
• Fits	• Meningitis
• Dementia	• Cranial nerve lesions
Pulmonary (10%–20%)	
• Chronic cough	• Pulmonary infiltrates
• Pleurisy	
Haematological (60%)	
• Anaemia	• Lymphadenopathy
Other (40%)	
• Fever	• Pigmentation

Investigations

Diagnosis is made by the characteristic features on small bowel biopsy, with characterisation of the bacillus by polymerase chain reaction (PCR).

Management

Whipple's disease is often fatal if untreated but responds well, at least initially, to intravenous ceftriaxone (2 g daily for 2 weeks), followed by oral co-trimoxazole for at least 1 year.

Symptoms usually resolve quickly and biopsy changes revert to normal in a few weeks. Long-term follow-up is Essential.

Bile Acid Diarrhea

A common scenario is in patients with Crohn's disease who have undergone ileal resection, which can also lead to other malabsorptive manifestations.

Unabsorbed bile salts pass into the colon, stimulating water and electrolyte secretion and causing diarrhoea.

If hepatic synthesis of new bile acids cannot keep pace with faecal losses, fat malabsorption occurs. Another consequence is the formation of lithogenic bile, leading to gallstones. Renal calculi, rich in oxalate, develop.

23.46 Classification of bile acid diarrhoea

Type	Cause
Type 1	Terminal ileal disease leading to malabsorption of bile acids from terminal ileum, e.g. Crohn's disease, terminal ileal resection
Type 2	Primary or idiopathic – no anatomical abnormalities or risk factors present
Type 3	Other conditions that affect GI absorption, e.g. chronic pancreatitis, coeliac disease, small intestinal bacterial overgrowth

• An elevated serum 7α-hydroxycholestenone is a useful non-invasive marker of bile acid diarrhoea.

• Diarrhea usually responds well to cholestyramine, a resin which binds bile salts in the intestinal lumen.

Short Bowel Syndrome (Intestinal Failure)

Intestinal failure (IF) is defined as a reduction in the function of the gut below the minimum necessary for the absorption of macronutrients and/or water and electrolytes such that intravenous supplementation is required to support health and/or growth.

19.21 Causes of short bowel syndrome in adults

- Mesenteric ischaemia
- Post-operative complications
- Crohn's disease
- Trauma
- Neoplasia
- Radiation enteritis

22.23 Management of short bowel patients (and 'high-output' stoma)

Accurate charting of fluid intake and losses

- Vital: oral intake determines stool volume and should be *restricted* rather than encouraged

Dehydration and hyponatraemia

- Must first be corrected intravenously to restore circulating volume and reduce thirst
- Stool volume should be minimised and any ongoing fluid imbalance between oral intake and stool losses replenished intravenously

Measures to reduce stool volume losses

- *Restrict* oral fluid intake to ≤500 mL/24 hrs
- Give a further 1000 mL oral fluid as oral rehydration solution containing 90–120 mmol Na/L
- Slow intestinal transit (to maximise opportunities for absorption):
  - Loperamide, codeine phosphate
- Reduce volume of intestinal secretions:
  - Gastric acid: omeprazole 20 mg/day orally
  - Other secretions: octreotide 50–100 μg 3 times daily by subcutaneous injection

Measures to increase absorption

- Teduglutide (a recombinant glucagon-like peptide 2) significantly reduces requirements for intravenous fluid and nutritional support by promoting growth and integrity of small bowel mucosa

Radiation Enteritis and Proctocolitis

Intestinal damage occurs in 10%–15% of patients undergoing radiotherapy for abdominal or pelvic malignancy.

The risk varies with **total dose**, **dosing schedule** and **the use of concomitant chemotherapy**.

The rectum, sigmoid colon and terminal ileum are most frequently involved.

Radiation causes acute inflammation, shortening of villi, oedema and crypt abscess formation, on the long term it can lead to fibrosis & adhesions.

Clinical Features

Acute : In the acute phase, there is nausea, vomiting, cramping abdominal pain and diarrhoea. When the rectum and colon are involved, rectal mucus, bleeding and tenesmus occur.

Chronic :The chronic phase develops after 3 or more months in some patients & usually presents as complication.

21.47 Chronic complications of intestinal irradiation

- Proctocolitis
- Bleeding from telangiectasia
- Small bowel strictures
- Fistulae: rectovaginal, colovesical, enterocolic
- Adhesions
- Malabsorption: bacterial overgrowth, bile acid diarrhoea (ileal damage)

Investigations

Management

1. **Diarrhea** :in the acute phase should be treated with codeine phosphate, diphenoxylate or loperamide.

2. **Antibiotics**: when there is secondary bacterial overgrowth.

3. **Nutritional supplements** : when malabsorption is present.

4. **Cholestyramine** : if there is associated bile acid diarrhea.

5. **Surgery** : Surgery should be avoided, if possible, because the injured intestine is difficult to resect and anastomose, but may be necessary if there is obstruction, perforation or fistula.

Note : steroids enema is usually NOT effective ! Why ?

Protein-Losing Enteropathy

It is an excessive loss of protein into the gut lumen, sufficient to cause hypoproteinaemia, **in the absence of other causes of protein loss** (**like liver diseases** & **proteinuria**).

22.58 Causes of protein-losing enteropathy

With mucosal erosions or ulceration	
• Crohn's disease • Ulcerative colitis • Radiation damage	• Oesophageal, gastric or colonic cancer • Lymphoma
Without mucosal erosions or ulceration	
• Ménétrier's disease • Bacterial overgrowth • Coeliac disease • Tropical sprue	• Eosinophilic gastroenteritis • Systemic lupus erythematosus
With lymphatic obstruction	
• Intestinal lymphangiectasia • Constrictive pericarditis	• Lymphoma • Whipple's disease

The **diagnosis** can be confirmed by measurement of faecal clearance of α1- antitrypsin or 51Cr-labelled albumin .

Treatment is that of the underlying disorder, with nutritional support and measures to control peripheral oedema.

Lactose Intolerance

There is genetically determined (primary) lactase deficiency (jejunal morphology is normal).

'Secondary' lactase deficiency occurs as a consequence of disorders that damage the jejunal mucosa, such as coeliac disease and viral gastroenteritis.

Clinical Features

In most people, lactase deficiency is completely **asymptomatic**.

Colicky pain, abdominal distension, increased flatus, borborygmi and **diarrhoea after ingesting milk or milk products** .

DX .: clinical improvement on lactose withdrawal & The lactose hydrogen breath test is a useful non-invasive investigation.

Treatment :

A. Dietary exclusion of lactose is recommended.

B. Addition of commercial lactase preparations to milk.

References:

Davidson's principles and Practice of medicine, 24rd edition , 2023.

لا تنسوا أهلنا في فلسطين و السودان من صادق دعائكم